

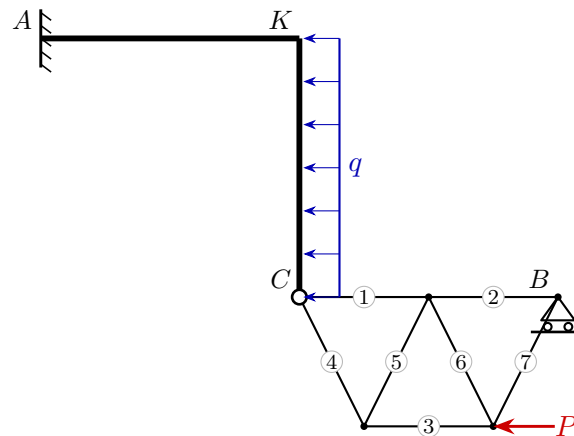
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Exam **Group A**

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

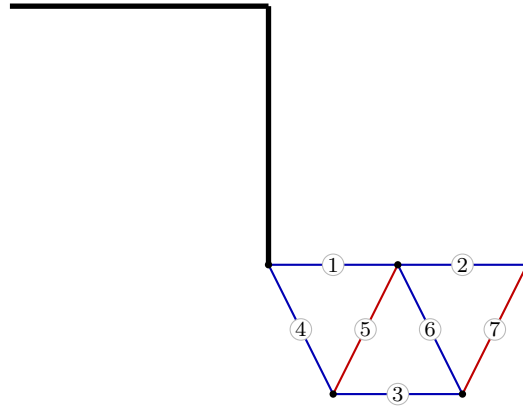
$A : R_x = 23, R_y = -7.5, M = 46$ $B : R_x = 0, R_y = 7.5$

$C : H_x = -15, H_y = 7.5$

3. Truss member forces

Member	Force S [kN]	Type
1	-11.25	compression
2	-3.75	compression
3	-7.5	compression
4	-8.39	compression
5	8.39	tension
6	-8.39	compression
7	8.39	tension

red: tension (+) blue: compression (–)



4. Section-force diagrams of the beams

